

E-Commerce APIs – Project Requirements

Overview:

The E-Commerce API is a backend system that allows users to:

- browse products.
- manage their cart.
- place orders.

The system supports:

- User authentication and authorization
- Product browsing with filtering and pagination
- Cart management
- Order processing

Architecture & Technical Requirements:

The project must follow:

- N-Tier Architecture (3 Layers + Common)
- Repository Pattern (Generic & Non-Generic)
- Unit of Work
- DTOs
- Fluent Validation
- Async Programming
- Result Pattern (General Response Wrapper)
- JWT Authentication
- Policy-Based Authorization
- CORS Enabled

Authentication & Authorization

- Use JWT Authentication
- Use Policy-Based Authorization
- Use Microsoft Identity
- UserId must be extracted from **JWT Claims** (not passed in request)

Main Components:

1. Users
2. Products
3. Categories
4. Cart
5. Orders

Main Functions and API Endpoints:

User Management:

- Register User: Create a new user account.
POST /api/auth/register
- Login User: Authenticate user and provide a token.
POST /api/auth/login

Category Management

- Get All Categories
GET /api/categories
- Get Category Details
GET /api/categories/{id}
- Create Category
POST /api/categories
- Update Category
PUT /api/categories/{id}
- Delete Category
DELETE /api/categories/{id}

Product Management:

- Get Products (Filter + Search + Pagination)
GET /api/products?categoryId=&name=&pageNumber=&pageSize=
- Get Product Details
GET /api/products/:id
- Create Product
POST /api/products
- Update Product
PUT /api/products/:id
- Delete Product
DELETE /api/products/:id

Cart Management:

- Add to Cart
POST /api/cart
- Remove from Cart
DELETE /api/cart/{productId}
- Update Cart Item Quantity
PUT /api/cart
- Get User Cart
GET /api/cart

Order Management:

- Place Order
POST /api/orders
- View Orders History
GET /api/orders
- Get Order Details
GET /api/orders/{id}

File Management:

- Upload Image:
POST /api/image/upload
- Product Image
POST /api/products/:id/image
- Category Image
POST /api/categories/:id/image

Important Notes:

- Use EF Core
- Backend only — No UI required
- Do NOT pass UserId in requests
- Extract UserId from JWT Claims
- Follow clean architecture principles

Delivery Requirements:

- Upload project to GitHub
- Ensure all source code is included
- Write a clean README file
- Record a Postman testing video and add the link in README